

experience. The following is a brief on the various types of display technologies in use for mobile phones.

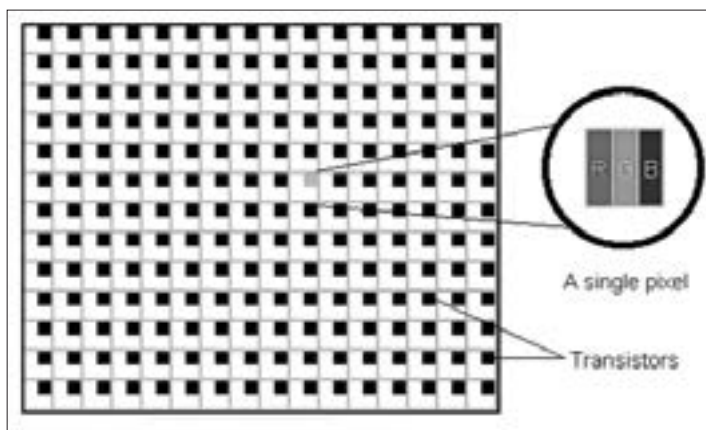
LCDs

LCD displays utilise two sheets of polarising material with a liquid crystal solution between them. Crystals in this suspension are naturally aligned parallel with one another, allowing light to pass through the panel. When electric current is applied, the crystals change orientation and block light instead of allowing it to pass through, turning the crystal region dark.

There are two main types of Liquid Crystal Displays: STN and TFT.

STN

Super Twisted Nematic LCDs use the passive matrix screen technology, which has no active or controlling element inside the display cell. Pixels are controlled by energising the appropriate row and column drive lines of the matrix from outside the display, resulting in a slow frame rate. STN screens have limited colour range and viewing angles (~ 15 degrees max).



Structure of a TFT LCD